



Stein's 'Regional City' Concept Revisited: Critical Mass and Complementarity in Contemporary Urban Networks

Author(s): Evert Meijers

Source: *The Town Planning Review*, Vol. 79, No. 5 (2008), pp. 485-506

Published by: [Liverpool University Press](#)

Stable URL: <http://www.jstor.org/stable/40112779>

Accessed: 28/06/2014 14:36

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at

<http://www.jstor.org/page/info/about/policies/terms.jsp>

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.



Liverpool University Press is collaborating with JSTOR to digitize, preserve and extend access to *The Town Planning Review*.

<http://www.jstor.org>

Stein's 'Regional City' concept revisited

Critical mass and complementarity in contemporary urban networks

Many decades after its conception, Stein's famous concept of the Regional City appears to be more alive than ever before. Unwittingly, the integrated development of a cluster of close-by cities has become a core issue in national and regional strategic planning, although the label 'urban network' is more common than 'Regional City' for such a cluster. In terms of ideal spatial organisation, both concepts are similar. The main assumptions in this respect are that the support base or critical mass in a polycentric cluster of cities is equivalent to that of a similar-sized monocentric city. Also, these close-by cities are assumed to complement each other in terms of urban functions. This article explores whether these assumptions hold true. It was found that for Dutch *Wet Gemeenschappelijke Regelingen* (WGR) regions the polycentric ones provide less support for amenities than monocentric regions, and that this was true even for those regions that are more complementary. The implications of these findings for the regional planning of such a cluster of cities are discussed.

The Regional City

The famous American architect and planner Clarence Stein devoted many years of his life to promoting the Regional City concept, a communitarian regional ideal. This concept, he argued, was the only appropriate remedy to the challenges of the post-war era, including rapid (sub)urbanisation, the need to foster local identity or a 'sense of belonging' as well as the need for a balance between urbanisation and environment. His ideal pattern for the city was a regional model, emphasising the relations between a constellation of communities, separated by great open areas but bound closely together by townless highways (see Figure 1).

Stein summarised his ideal pattern for the building and redevelopment of future and existing metropolises in a 1964 article in *Architectural Record*. The different urban areas of a Regional City

will have varied functions, character, forms and sizes. Each will be large enough to support all the day-to-day activities and equipment of a contemporary American community. All towns will be small enough to foster local loyalty and pride, as well as closeness to country-side. The total population of a regional group of towns and farms will be adequate to utilize and support central facilities equivalent to those provided by

Evert Meijers is a researcher in the OTB Research Institute for Housing, Urban and Mobility Studies, Delft University of Technology, PO Box 5030, 2600 GA Delft, The Netherlands; email: e.j.meijers@tudelft.nl

Paper submitted December 2007; revised paper received and accepted July 2008.

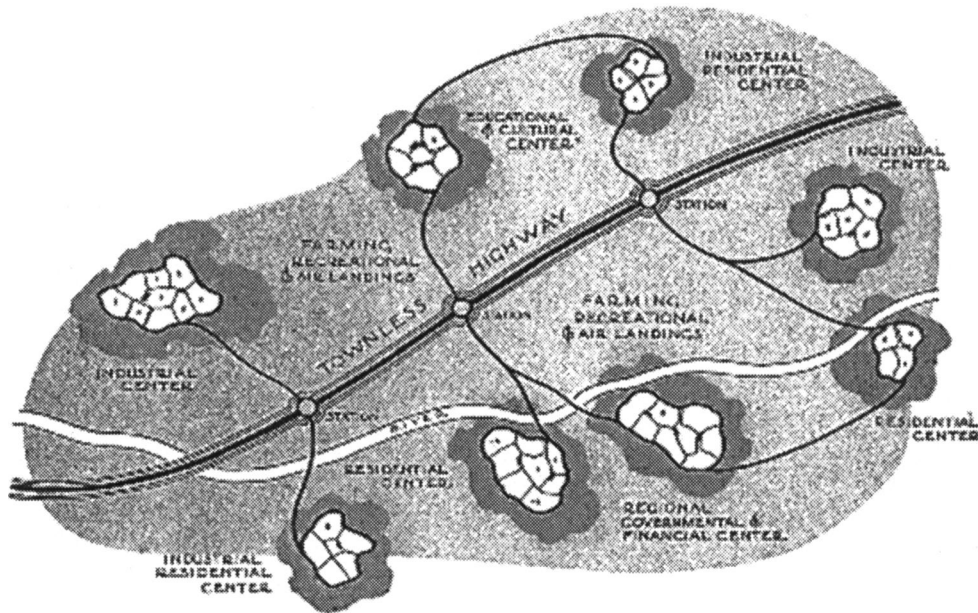


Figure 1 Stein's Regional City
Source: Stein (1942)

a great metropolis of today ... These main central facilities will be in various towns, each of which will serve one or more at most a few related functions. Thus traffic loads will be limited and balanced. (Stein, 1964, 205–206)

Stein worked on a variety of towns, but his blueprint for the Regional City, meant to replace the famous Garden City concept on which it builds, was never wholly realised.

A widespread policy concept

However, almost a century later – Stein and Mumford had been working on the Regional City concept since the 1920s (Parsons, 1998) – his ideas seem more alive than ever before. As the challenges faced today are still practically the same, the potential remedy also seems still to be the same. Unwittingly, his blueprint for the ideal metropolis has been embraced by a wide variety of national and regional planners and developers trying to develop what they often label ‘urban networks’: clusters of proximally located cities together allegedly providing urbanisation economies on the regional level. Such clusters of cities appear as planning concepts in planning and urban and regional development policies in a wide variety of European countries including

Belgium ('urban networks'), Denmark (polycentric 'National Centres'), Estonia ('urban networks'), France ('*réseaux des villes*'), Germany ('*Städtenetze*'; some of the '*Metropolregionen*'), Greece ('Twin poles'), Italy ('*reti di città*'), Ireland ('linked gateways'), Lithuania ('Metropolis Vilnius–Kaunas'), the Netherlands ('urban networks'), Poland ('*duopols*') and Switzerland ('*vernetzte Städtssystem*') (Meijers, 2005).

In England, awareness of the importance of focusing attention on city-regions has recently increased rapidly (Finch, 2007; Parkinson et al., 2006; HM Treasury et al., 2007). City-regions would be a more accurate proxy of the scale on which housing markets, travel-to-work areas and retail catchments function, and many policy issues can be more effectively addressed at this spatial scale (Marvin et al., 2006). Unlike several continental European countries, these city-regions are still defined as a core city and its hinterland, and not as a collection of core cities whose hinterlands overlap. Nevertheless, in several cases there is a need to do so, which becomes apparent from bottom-up attempts for instance in the Tees Valley city region, the Birmingham–Coventry–Black Country sub-region or the Three Cities sub-area composed of Nottingham, Leicester and Derby. Sub-regional planning for such polycentric regions is now fostered by the introduction of Multi Area Agreements (MAA) in the Local Government White Paper (DCLG, 2006) and so far 13 sub-regional areas have either recently developed them or are in the process of doing so. These MAAs provide a framework to address primarily strategic, economic development-related issues that span more than one local authority's or local strategic partnership's geographical boundary.

The unfolding of policies at the scale of polycentric city regions fits well with the trend of Western European national states replacing traditional redistributive spatial policies with new forms of locational policy that target strategic sub-national spaces as key sites for economic regeneration (Brenner, 2004).

'Regional Cities' of different sizes

Besides the policy debate, in academic literature contemporary versions of the Regional City concept have also recently come to the fore, for instance the 'city network' (Camagni, 1993) or 'network city' (Batten, 1995), while concepts such as the 'polycentric urban region' (Kloosterman and Musterd, 2001; Parr, 2004) or 'polynucleated metropolitan region' (Dieleman and Faludi, 1998) refer to the morphological characteristics of Regional Cities rather than the relationships between the cities constituting such regions. Classic and large examples of such regions include the Randstad in the Netherlands and the Ruhr Area in Germany, but the definition of a polycentric urban region is not size-specific: it is a collection of historically distinct and both administratively and politically independent cities located in close proximity, well connected through infrastructure and lacking one dominating city in political, economic, cultural

and other aspects (Kloosterman and Lambregts, 2001). Evidently, such clusters of cities can be found throughout Europe. While the academic debate has so far focused on the larger metropolitan regions (such as the Randstad or the RheinRuhr), polycentric development is also increasingly seen as a policy concept relevant for smaller and sometimes more remote areas; for instance, to guarantee access to a basic level of services and facilities in smaller, often more rural regions. This has become increasingly important in light of the 'Territorial Cohesion' objective, as can be found, for example, in the Territorial Agenda of the European Union. The European Spatial Development Perspective (ESDP; CEC, 1999) puts it as follows:

The creation of networks of smaller towns in less densely settled and economically weaker regions is also important. In these areas, co-operation between urban centres to develop functional complementarity may be the only possibility for achieving viable markets and maintaining economic institutions and services which could not be achieved by the towns on their own. (CEC, 1999, 21)

In this article I refer to such regions using the label 'urban network', not because this is necessarily the right label – all it implies is a polycentric cluster of cities, with no information as to whether and how they are 'networked' together – but because it is the most common concept applied to such regions in policy documents and my objective is to discuss the policy consequences of empirical analyses in such regions.

Much claimed, little proven

Obviously, there is a difference in terms of genesis between the concept of the urban network and Stein's Regional City. While a clustering of cities often evolved either from a history of fragmented political and administrative structures as well as rivalry (e.g. Randstad), or from rapid urbanisation during the Industrial Era following the discovery of natural resources as ore and coal (Ruhr Area), this polycentric pattern gained significance more recently as relative distances between them decreased and functional relationships started to increase. Conversely, according to Stein, a Regional City would result from a deliberate integral planning design. However, in terms of spatial structure characteristics, both concepts hold the same ideal configurations. Stein makes two main assumptions in this respect.

- (1) The mass or support present in the polycentric Regional City is equivalent to those generated by monocentric cities, which would result in a similar presence of urban facilities and amenities.
- (2) Facilities and amenities are dispersed across the constituent cities or towns in such a way that each one of them hosts some of them, but not the same, thus making the cities or towns complement one another.

Late contemporaries of Stein have provided circumstantial evidence for the actual existence of such urban networks, where the spatial structure complies with these assumptions (Burton, 1963; Gottmann, 1961). In contemporary policy documents also, administrators eagerly indicate the combined critical mass of their urban network, which would enable them to play in the 'major leagues' of city competition (Priemus, 1994). In addition, it is only too often that complementarities are emphasised in present-day policy documents. The following quote from a 2002 Dutch policy report by the Central Government that first introduced the policy of 'urban networks' in the Netherlands appears exemplary of the policy-maker's view on urban networks. According to this document, the strength of urban networks is that:

cities in networks do not duplicate each other, but exploit chances to complement each other and co-ordinate a certain division of labour. Amsterdam, Rotterdam, The Hague and Utrecht together with the other cities in the Delta Metropolis have much more to offer when taken together instead of individually and should be further developed in a coherent and co-ordinated way. (Ministerie van VROM, 2002, 32, author's translation)

Furthermore,

it is not the individual cities, but the metropolitan areas and urban networks that offer a complete range of living/working environments, services, parks and transport systems. (Ministerie van VROM, 2002, 33, English summary version)

This idea sets aside the previously dominant idea in Dutch planning of developing 'complete cities' (see Zonneveld and Verwest, 2005). A succeeding policy report, the National Spatial Strategy 'Nota Ruimte' (Ministerie van VROM, 2005), puts relatively more emphasis on the co-operative aspects of urban networks. It is expected that through co-operation in urban networks the spatial-functional structure of urban networks will change over time:

co-operation must enhance the competitive power of all cities in the network and ensure the presence of an attractive (international) location environment and a large diversity of urban functions. The cities and centres in the network complement and mutually enforce one another as together they are able to deliver more than when taken individually. The government expects the cities to co-ordinate tasks to allow for specialisation and complementarities. (Ministerie van VROM, 2005, 70)

Another good illustration of how vibrant Stein's ideas still are today can be seen in the spatial strategy for North Northamptonshire, England, adopted in June 2008, which is, according to the review of sub-national economic development and regeneration (HM Treasury et al., 2007), one of the first examples of sub-regional collaboration in the field of town planning. The region forms part of the East Midlands, a region that

lends itself to polycentric development policies (Coombes et al., 2005). In the spatial strategy for North Northamptonshire, which is composed of the proximally located cities of Corby, Kettering and Wellingborough, it is observed that, for an area its size, it fails to provide a number of important services and facilities, and as such loses growth to competing cities in neighbouring regions as people go there to fulfil their needs. Fostering the networks between the three cities should solve this situation:

The Plan provides a framework within which the component parts of North Northamptonshire can work closely together to secure more investment and a greater range of facilities and services than they could do by working alone. The Plan is based on increasing the self-sufficiency of individual settlements but also developing, where possible, a different focus for each in support of complementary working rather than wasteful competition. In this way North Northamptonshire as a whole can become stronger and more self sufficient, competing more effectively with other areas and stemming losses of people and spending. (North Northamptonshire Joint Planning Unit, 2008, 22)

Again, we can see Stein's two main issues – organising critical mass and developing complementarities – reflected in contemporary policies. In fact, these Dutch and English examples appear stereotypical of many territorial development strategies for urban networks that can be found across Europe.

This article explores the validity of Stein's two fundamental assumptions. Do these assumptions regarding the support base (or critical mass) and complementarity hold for its contemporary counterpart, urban networks? Planning for polycentric urban regions in many cases involves planning on a relatively new scale, based upon new starting points and taking on board new strategic objectives (Lambregts, 2000). Not much is known about effective spatial strategies for polycentric urban regions. Here, building on Stein, we examine the feasibility of two possible development objectives, namely organising critical mass and developing further complementarities between the cities.

The next section explores the assumption that the support base in an urban network for amenities is similar to a comparably sized monocentric single city. In the following section I shall explore the second assumption of complementarity. Empirical evidence relating to both assumptions was gathered for Dutch regions. Finally, I shall address the question of how utopian Stein's concept of the Regional City is, and whether its basic assumptions could and should be considered points of departure for planning modern-day urban networks.

Critical mass in urban networks

Many advanced or rare high-level urban amenities need a considerable minimum market size as regards both the demand for the amenities offered and the supply of

the necessary human capital. Therefore, a certain critical mass is deemed necessary for urban amenities to be able to diversify and function smoothly. Not surprisingly, the most highly valued economic activities and the widest variety in economic and cultural functions are found within the largest agglomerations. As Capello (2000, 1926) argues, 'the limit that the medium-sized cities come up against, and which often makes them succumb vis-à-vis the great metropolis, is the limit of critical mass and centrality'. It is a strategic issue whether such activities 'can also be realized in polynuclear urban structures, which lack the critical mass of large cities with agglomeration economies' (Lambooy, 1998, 459). Alonso (1973) presents in this respect the concept of 'borrowed size', which refers to the situation in which proximal and well-connected cities host urban functions normally found only in larger cities, such as top-level urban amenities, because the support base is larger given the proximity of more cities.

Not surprisingly, the idea of integrating formerly separate cities to form a larger entity able to reap the advantages of urban size is appealing to administrators and planners. However, the idea that the integration of separate cities results in agglomeration advantages comparable to similar-sized monocentric cities is 'rather simplistic' (Bailey and Turok, 2001). In this respect, Parr (2004) points to the need for longer travel flows, longer commodity flows and less convenient flows of information in urban networks. Moreover, it should be reckoned that 'some of the advantages of urban size stem from the nature of the metropolitan environment, and are related to such factors as density, proximity, face-to-face contact, informal structures, unplanned interaction, etc.' (Parr, 2004, 236), and consequently hold less for urban networks. Moreover, in this respect, historic development pathways do not favour the position of urban networks either. Cultural, leisure and sports amenities tend to be strongly rooted in their urban context, and their presence in qualitative and quantitative terms is strongly based on the city and the size of its population, including that of its 'rural' hinterland. Simply pooling the urban amenities present in the different cities of an urban network together is likely to lead to duplication rather than to a higher level of the urban amenities at stake. For instance, having three medium-sized theatres in three medium-sized cities together constituting an urban network does not mean they can attract those top-end theatre productions that demand more physical space as well as larger receipts and thus more attendants. Such productions tend to be reserved for larger theatres, generally found in larger cities. In sum, many doubts can be raised concerning the support base, or critical mass of Regional Cities/urban networks in comparison to similar-sized monocentric urban regions.

A recent research project (Meijers, 2008) empirically explored this question of whether a polycentric spatial layout does influence the provision of cultural, leisure and sports amenities in urban networks indeed. Forty-two Dutch regions that were delimited according to the Intermunicipal Statutory Regulations Act (*Wet Gemeenschappelijke*

Regelingen, WGR), an indirect proxy for functionally coherent regions,¹ served as cases. This WGR act is very similar to the recent Multi Areas Agreement in England, and as a consequence, WGR regions can be considered equivalent to the English city-regions for which an MAA has been set up. Their average population was 387,000 and it is generally possible to move between any two places in such a region within half an hour maximum. For these regions, an inventory was made of the presence in these regions of cultural, leisure and sports amenities with a potentially supralocal or regional significance, such as cinemas, art house cinemas, theatres, museums, concert halls, pop/concert venues, casinos, Michelin-star restaurants, indoor ski-halls, skating rinks, climbing halls and sports stadiums. The extent to which these amenities were present in each region was summarised in an index² that served as the dependent variable in a multiple regression model. Explanatory variables in the model included the population size of the region; the number of visitors (tourists, business travellers) to the region, for which the average number of occupied hotel beds in the region served as a proxy; the average household income in the region; and the score of a region on a scale ranging from very monocentric to very polycentric.

This latter score was based on the rank-size distribution of cities within a region (see Nordregio et al., 2004). Characteristic for an urban network is that no city dominates the others. In other words, an urban network lacks a clear hierarchy. The rank-size distribution of the regional system provides information on this hierarchy and is therefore an excellent indication of the extent of mono- or polycentricity. The more even the sizes of cities within a region are distributed, the more polycentric.

- 1 Ideally, the case study regions would be delimited on the basis of functional relationships relating to the use of cultural, leisure and sports amenities. However, in the Netherlands, such detailed travel data is not sufficiently reliable. Instead, use is made of a regional delimitation that is relatively recent, that is not by definition confined to old provincial administrative borders and that provides for a proxy of a functionally coherent region: the *WGR-samenwerkingsgebieden*. These regions have been delimited within the framework of the Intermunicipal Statutory Regulations Act (*Wet Gemeenschappelijke Regelingen*). This act enables municipalities to work jointly on issues that need to be addressed on a higher spatial scale than the municipal scale by means of issue-based common arrangements. The Act does not specify which issues should be jointly addressed, but in practice these often concern regional aspects of economic development, tourism, recreation, housing, employment, traffic and transport, spatial development, nature and environmental affairs, welfare and social affairs. Until recently, the act required that such common arrangements had to be clustered within the same regions in order to avoid a patchwork of common arrangements that was too territorially uncoordinated. As the delimitation of WGR regions is based on the municipal and provincial administrators' and councillors' perceptions of the scale on which issues in need of a regionally coordinated approach rise, these regions provide an indirect proxy of functionally coherent regions.
- 2 Frequency data for 20 different types of amenities was collected and grouped into six categories: cinema, theatre, museum, concert, sports and a rest category comprising leisure activities not yet belonging to the other categories. The scores of the 42 WGR regions on these 20 variables were normalised and I used these z-scores to develop my index. Each amenity within a category (cinema, theatre, etc.) was given the same weight, and each of the six categories was also given the same weight in the total index. Therefore, the total index is the average score of the six constituent categories.

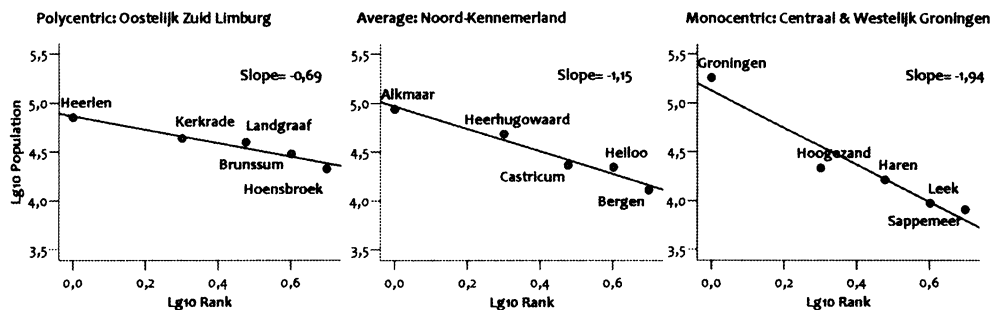


Figure 2 Monocentric and polycentric rank-sized distributions based on five towns per region

Figure 2 presents the size distribution of the five most populous cities or towns in three WGR regions, as well as the regression line that best fits this distribution. The more flat this line is, the more polycentric the region, and vice versa. Therefore, the slope of this regression line is our indicator of the extent of mono- or polycentricity, used in the multiple regression model.

The regression analysis showed that all four independent variables contribute significantly to explaining the presence of cultural, leisure and sports amenities, while the model itself has high explanatory power (adjusted $R^2 = 0.948$). This is particularly due to the population size factor (Beta = 0.761), followed at some distance by the number of visitors (Beta = 0.264). Given the perspective of the present article, the most important finding is clearly that the extent of mono- or polycentricity is also of significance for explaining the presence of these amenities (Beta = -0.109, level of significance 0.01).³ This relation is negative, which means that the more polycentric a region is, the fewer such amenities are present, or, put another way, the more monocentric, the more cultural, leisure and sports facilities it has. So, when correcting for differences between WGR regions in terms of population, the number of visitors and the average income, clusters of cities have a disadvantage compared to single cities: the more polycentric a region, the less cultural, leisure and sports amenities are present. In other words, five cities of, say, 100,000 inhabitants, located close by, cannot organise the same support base or critical mass as one single city of 500,000 inhabitants. Therefore, Stein's assumption that '[t]he total population of a regional group of towns and farms will be adequate to utilize and support central facilities equivalent to those provided by a great metropolis of today' is refuted. The specific dispersed spatial layout of polycentric regions implies that they cannot reap the advantages of

- 3 In order to check for spatial autocorrelation, a fifth explanatory variable was included in the model. This was the average score on the index of cultural, leisure and sports amenities of the regions directly bordering the WGR region. This fifth explanatory factor is far from significant (Sig. = 0.89) and does not change the findings for the other variables. There is no spatial autocorrelation.

urban size in the same way as more monocentric urban regions, at least, as far as our selection of cultural, leisure and sports amenities is concerned.

Complementarity is a means through which more critical mass could be achieved in an urban network. This issue is discussed in the next section.

Complementarity in urban networks

Complementarity between cities is another major assumption raised by Stein. According to Stein, in a Regional City facilities and amenities would be dispersed across cities or towns in such a way that each one of them hosts some of them, but not the same, thus making the cities or towns complement one another.

So, complementarity refers to the idea that different cities fulfil different and mutually beneficial roles (Hague and Kirk, 2003), for instance through providing different sets of urban facilities and amenities. Complementarity results from the differentiation between centres or cities in terms of urban functions, while these urban functions should be provided for, at least partly, the same geographical demand market (Meijers, 2005; Ullman, 1956; Lambooy, 1969).

Using correspondence analysis techniques (see for instance Greenacre, 1993, for methodological details), it is possible to calculate and plot the division of labour in economic activities between the cities making up an urban network. Using this method, it has been shown (Meijers, 2007a) that a strong division of labour exists between the cities making up the Randstad (Amsterdam–Rotterdam–The Hague–Utrecht). As far as service sector activities are concerned, its cities are for instance twice as specialised in comparison to each other than the cities in the German Rhein-Ruhr Area (Cologne–Düsseldorf–Essen–Dortmund). In addition, it was found that the extent to which cities complement one another in the Randstad, as well as in the Flemish Diamond (Brussels–Antwerp–Ghent–Leuven; see Albrechts, 2001) decreased rapidly in the 1996–2002 period. As this analysis used rather general categories of service sector activities (in fact, the two-digit NACE rev. 1.1 classification), little is known of complementarity and trends using a much more detailed coding of economic activities.

Here, I present an analysis using a very detailed classification of economic activities (NACE rev. 1.1 four-digit codes) that allows us to examine the division of labour between cities making up an urban network in terms of the type of urban amenities with a (potential) supralocal significance that Stein referred to in his work on the Regional City: theatres, museums, restaurants and bars, department stores, schools, hospitals, sports accommodations, hotels, cinemas and a variety of commercial services such as banks, insurance agents and employment agencies (see Appendix 1 for the complete list of the 25 types of amenities included in the analysis). The six WGR regions that turned out to be very polycentric on the basis of their rank-



Figure 3 Polycentric WGR regions

Table 1 Complementarity ratios for urban amenities in polycentric regions measured in terms of employment

Region	Employment			Trends (%)		
	1996	2001	2006	96–01	01–06	96–06
Oost-Groningen	0.147	0.155	0.171	6	10	17
Noord-Groningen en Eemsmond	0.204	0.194	0.176	–5	–9	–14
Twente	0.120	0.117	0.120	–3	2	–1
Eem en Vallei	0.064	0.072	0.064	13	–12	–1
Noordwest-Veluwe	0.137	0.123	0.126	–11	3	–8
Oostelijk Zuid-Limburg	0.057	0.065	0.072	13	11	26

size distribution⁴ served as cases. These regions are depicted in Figure 3. Data on the presence of urban amenities were collected for the three major cities per WGR region, at the spatial scale of municipalities. For each of the 25 detailed classes of urban amenities, data on employment (fte) in each of the cities was gathered from the National Information System on Employment (LISA) database, for the years 1996, 2001 and 2006.

Table 1 presents the extent to which cities within each region complement one another in terms of the selection of urban amenities (theatres, museums, restaurants and bars, department stores, schools, hospitals, sports accommodations, hotels, cinemas and a variety of commercial services such as banks, insurance agents and employment agencies). These scores are based on the total inertia, which is the Chi-square statistic divided by n of the contingency tables with the cities in the rows and the economic activities in the columns which were derived from the LISA Database. The complementarity ratio is calculated through normalising the total inertia by dividing it by the maximum total inertia possible, which is equal to the dimensionality of the problem (in our case, the number of cities – 1). This results in scores between 0 and 1. A score of 1 would be reached if each of the three cities in a polycentric region hosts a completely different set of amenities, for instance all banks in city A, all theatres in city B and all restaurants in city C. This is what Stein envisaged to be the case in a Regional City. A score of 0 is attained when all types of amenities are equally spread over the three cities according to their size (measured in terms of total number of jobs). In that case, cities A, B and C will all host the same types of urban amenities. In other words, a score of 0 represents maximum duplication and no complementarity between the cities.

At this very detailed level of analysis of economic activities, we find that there

4 Those regions with a score of more than 1 standard deviation above the mean of the scores on the monocentric–polycentric scale, as indicated by the slope of the regression line through its rank-size distribution (regression line calculated for three cities per WGR region).

are large differences between the six polycentric regions examined. In 2006, in the region with the highest score (Noord-Groningen en Eemsmond), the cities were 2.75 times more complementary than the region with the least complementary city network (Eem en Vallei). Still, the extent of complementarity is substantially higher than was the case for complementarities in terms of service sector activities (measured at the two-digit level of detail) between the main cities of the Randstad (2002, 0.042), Flemish Diamond (0.033) or RheinRuhr (0.022) (see Meijers, 2007a). Therefore, it appears that in general, complementarities are quite high in these six regions.

Trends in this extent of complementarity do not show a robust pattern. In some regions, complementarities increased considerably (Oost-Groningen, Oostelijk Zuid-Limburg), in others complementarities remained the same (Twente, Eem en Vallei), while in other regions complementarities decreased (Noord-Groningen en Eemsmond, Noordwest-Veluwe). The average extent of complementarity for all regions together (0.121) remained the same for 1996, 2001 and 2006. The locational decisions of local firms and amenities account for these changes, as does their growth and decline.

One of the main advantages of correspondence analysis is that it graphically displays the associations between the cities, thus enabling an easier interpretation of how the amenities of one city relate to the amenities of another. As an example, Figure 4 presents a picture of how complementarities between the cities in two regions (Oost-Groningen and Twente, respectively) developed over the 1996–2001–2006 period.

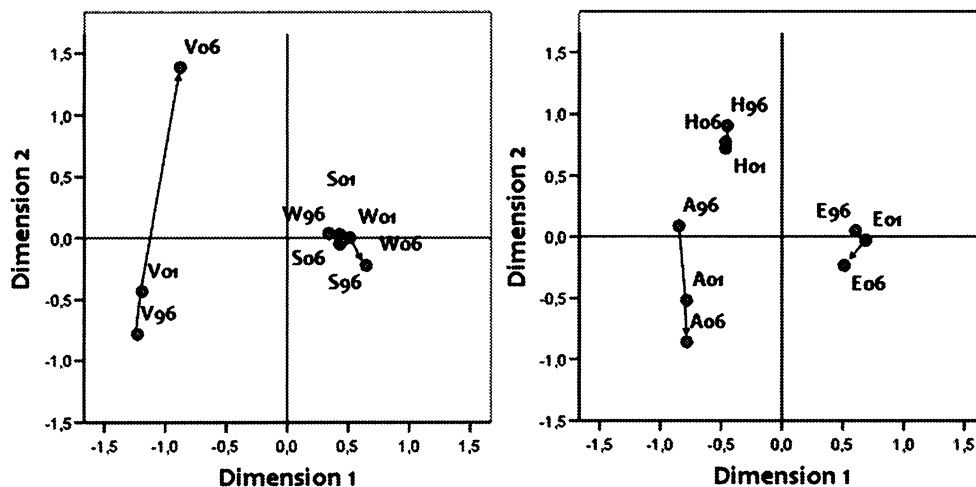


Figure 4 Development in complementarities between cities in Oost-Groningen (left) and Twente (right) over the 1996–2006 period. Oost-Groningen: S, Stadskanaal; V, Veendam; W, Winschoten. Twente: A, Almelo; E, Enschede; H, Hengelo.

This plot should be read as follows. In the plot, two axes together indicate the origin (0,0), which resembles the average set of amenities of the cities in one region for the years 1996, 2001 and 2006 simultaneously. The further a city is away from the origin, the more it contributes to the extent of complementarity. In addition, if two cities lie close together in the plot, then their urban amenities are more or less similar. If the distance between two cities increases, then they have become more complementary. Conversely, when two cities move in each other's direction, then they are becoming more similar in terms of urban amenities. The position of the cities in each plot given by its first character, followed by the year concerned, e.g. Eo6 is the position of the city of Enschede in 2006.

What Figure 4 clearly expresses is that amenities can be distributed over a region's cities in very different ways, ways that do not emerge solely from looking at the overall complementarity ratio. For instance, the cities in Twente (on the right) are all quite distinct from each other, while in Oost-Groningen (left) it is obvious that the cities of Winschoten and Stadskanaal are very similar in terms of the types of amenities they host, with Veendam providing different types of amenities and becoming even more dissimilar from the other cities. The trajectories of cities in the plots of Figure 4 provide for even more detail, for instance that in terms of urban amenities the city of Enschede is moving towards the profile of Almelo, which in turn is becoming more dissimilar from that of Hengelo.

Complementarity to gain critical mass?

An intriguing question is whether the regions that are more complementary also score better in terms of critical mass. This is the essence of Stein's ideas on the Regional City and also comes forward in contemporary strategic development policies for urban networks, namely the assumption that it is exactly through the fact that cities complement one another that they achieve more critical mass.

The present analysis may shed some light on this question, although the limited number of regions in the analysis of complementarity does not allow for definite answers. In the analysis of critical mass, it was established that the size of the residential population is by far the most important factor for explaining the presence of cultural, leisure and sports amenities. In order to correct for the population size of regions all the values of the variables expressing frequencies of cultural, leisure and sports amenities (those amenities analysed in Section 2) were made relative to the population number, resulting for instance in the number of seats in cinemas per capita, the number of theatres per capita, the number of various sports amenities per capita, etc. Correlating these relative frequencies with the extent of complementarity in each region, however, revealed hardly any significant correlation. Exceptions are museums and the number of karting tracks per capita, which were found significantly

more often in more complementary regions. However, given the large number of variables for which there is no correlation at all, these appear to be the exceptions to the rule. This has implications for planning strategies in urban networks, on which the next section elaborates.

Planning urban networks

In many European countries, the integrated development of urban networks has become a core issue in national and regional strategic planning. Many claims are made for such urban networks in regional development strategies. Those relating to the spatial organisation of such regions include the claim that the support base or critical mass in an urban network is equivalent to the support base in a single metropolis. Another claim is that cities complement one another in urban networks. These claims resemble very much Clarence Stein's ideas of what he termed the Regional City. The question then arises of how utopian Stein's ideas were. The fact that the concept of a Regional City, albeit under a different label, has become so popular decades later does not make it utopian. However, according to the empirical evidence presented in this article, these claims or assumptions with respect to the spatial organisation of a network of cities need to be qualified.

- The idea that the critical mass or support base for urban amenities and facilities in a network of cities is equivalent to those in a similar-sized single city does not hold true for our case of Dutch WGR regions, with respect to cultural, leisure and sports amenities. For these types of amenities, significantly less support is generated in polycentric regions.
- Although there may be quite a strong division of labour between cities, this is on average not increasing and in some cases, when analysed on a less detailed level, is even decreasing.
- So far, no indication was found that the regions in which the cities are more complementary are also characterised by a greater critical mass. However, this requires more research as our sample was rather small.

The question then is how planners should deal with these outcomes and what consequences they have for planning urban networks. Here, I shall discuss two issues. First, I shall consider the way complementarity and a regionally based critical mass come about. Second, I address the way planning can contribute to bringing about complementarity and critical mass. Finally, I reflect on the appropriateness of enhancing complementarity and critical mass as policy objectives.

The coming about of complementarity and critical mass

One of the first issues is how a regional support base and complementarity develop. To start with, it appears that complementarity and critical mass mutually reinforce one another. When cities complement one another, it requires actors in the shared hinterland to develop a regional orientation in order to fulfil their demand for more specialised services or products. This regional orientation then fosters a larger critical mass. A larger, regional support base also allows for further specialisation. When these specialisations develop in different cities, it follows that complementarity rises.

Urban development pathways are important in explaining the genesis of complementarity and critical mass. In that sense, there seems to be a difference between urban networks that have been shaped over the past centuries as a result of fragmented political and administrative structures and rivalry, thus preventing the rise of one continuously dominant city, and, on the other hand, urban networks that developed from rapid urbanisation during the Industrial Era following the discovery of natural resources such as ore and coal. It appears that complementarities in the latter type of region (e.g. RheinRuhr, Oostelijk Zuid-Limburg) are less high developed than in the former type of urban network (see also Meijers, 2007a). Another factor is the extent to which the cities in an urban network compete with one another. In many cases, it seems common for the cities in an urban network to share a long history of rivalry and competition. It has been suggested that interurban competition leads to duplication rather than complementarity: 'The most oft-noted drawback of inter-territorial competition is serial reproduction, the imitation and replication of the same ideas from place to place' (Malecki, 2004, 1112). Cities tend to market the same kind of images (Holcomb, 1994) and copy the same forms of urban redevelopment, innovations and investments that have already been proven to make particular cities more attractive (Harvey, 1989). In other words, strong competition tends to lead to duplication rather than complementarity.

In Meijers (2007b) a detailed account of the development of complementarity in the hospital and higher education sectors is provided. It led to the observation that whether or not complementarity develops depends on a complex balancing of macro-level conditions such as laws, rules, central government funding schemes and the micro-level behaviour of organisations, individuals or households, that is their behavioural rationales following from the pursuit of their interests. These interests generally follow a market principle: private actors aim to optimise their own interests within the market conditions set at the macro-level. Public actors influence only part of these macro-level conditions, but may still have a significant impact. Therefore, whether or not complementarity develops seems the result of a series of coincidences and is highly dependent on the complex interplay between macro-level conditions and micro-level rationales. As regards the latter, actors on this micro-level are often

not within the realm of public administration, have different agendas and are not committed to planning objectives such as enhancing complementarity.

Planning for complementarity and critical mass

A second issue is how planning could contribute to increasing complementarity and critical mass. Planning for such regions in many cases involves planning on a relatively new scale, based upon new starting points and taking on board new strategic objectives (Lambregts, 2000). Here, we present some deliberations on what appear to be the major issues, or starting points, that may enable a successful regional planning strategy in this respect.

The best possibilities for purposefully encouraging complementarity in urban networks are likely to occur when new publicly funded supralocal amenities, facilities and business locations are developed simultaneously. However, even in such cases, the elaboration of development plans has proven to be very difficult. A case in point is the '*Nadere Uitwerking Randstad Internationaal*' (NURI), a programme drawn up in the early 1990s under the guidance of the administrators of the Randstad provinces and major cities. The ambition of this programme was to arrive at a certain functional division of labour between the main city regions in the Randstad in order to enhance its metropolitan nature through, amongst others, developing high-order international-class amenities and avoiding duplication and an impending surplus of business locations and office space. During the process of developing this programme, it became obvious very quickly that such an enhancement of complementarity at this scale and at that time was unmentionable and appeared impossible (Zonneveld, 1992; Zonneveld and Verwest, 2005). A complicating factor with respect to the spatial allocation of facilities and amenities is that their attractiveness varies: for instance, some will have a NIMBY character, while there is strong competition for others. In addition, the temporal dimension adds complications, as benefits and costs do not accrue to the same actors, nor do they come up at the same time.

Therefore, a precondition for regionally coordinating the spatial allocation of amenities, facilities and business locations is the emergence of regional organising capacity. This capacity, to be understood as a more or less institutionalised regional framework of cooperation, debate, negotiation and decision-making on spatial development (Meijers and Romein, 2003) should also include mechanisms to reconcile different interests and balance benefits and costs, even when these do not appear simultaneously. Coordination between the cities should prevent them from developing the same kind of projects (e.g. renewal of their station areas) or trying to attract the same kind of companies (ICT, business services). It appears that such duplication is also increasingly economically inefficient when both the functional integration and the internal accessibility within a region further increases.

Regional organising capacity is also needed for what could be considered the most important step towards an enhanced critical mass and complementarity in urban networks: the shift from local to regional funding of amenities and facilities and from a local tax base towards a regional tax base. This would make regional coordination and cooperation much less informal and free of obligations. If central government resources accrued more to regional rather than local entities, and were based on regional size, rather than local size, this would also prevent the discrimination of urban networks *vis-à-vis* monocentric regions having a dominant urban core, because, at present, it is in many countries the case that larger cities tend to receive disproportionately more central government funding. The latter can be considered a major explanation for our finding that the polycentric regions in the Netherlands are outperformed by monocentric regions in terms of the presence of cultural, leisure and sports amenities.

This could all be stimulated by an increased awareness in urban networks of the need for complementarity and critical mass. This could be achieved through benchmarking urban networks on both issues, and also in relation to the economic and social performance of the regions. In addition, monitoring the gap with similar-sized single cities would similarly foster this 'regional awareness'. Finally, increasing information on urban activities and possibilities in the network could lead to a different, regional rather than local mindset for both the inhabitants and companies. Regional symbols could also enhance a regional identity.

Obviously, the development of transportation systems on the scale of the network of cities, as well as knitting the region more strongly together through its infrastructures would enhance complementarity and critical mass as well, in line with Stein's 'townless highways'.

Reflection

So far, it has been taken more or less for granted that Stein's vision of complementarity and an enhanced critical mass are appropriate policy objectives for planning urban networks. Arguments underpinning this statement can indeed be provided, but the question is more complex given the relationships between critical mass and complementarity.

The benefits of complementarity and critical mass for firms and households appear self-evident: it enables a more specialised and diversified array of urban functions from which to choose from. However, one needs to bear in mind that critical mass and complementarity are not of the same order. Enhancing critical mass should be considered the main challenge for planning in urban networks. This will directly raise the extent of specialisation, which can relate to urban facilities, amenities, business services, residential environments, business locations, etc. Thus, the idea is that a 'top' level should be added, reflecting the support base present in the urban network rather

than at the local level of its constituent cities. Such specialised activities or environments can be distributed rather evenly across the cities of the urban network, which implies a situation of complementarity. Alternatively, however, these can also be more or less clustered or concentrated in one and the same city. In both cases, the region profits from increased specialisation, but the spatial pattern is very different. Complementarity is therefore just one of two modes through which an enhanced critical mass takes shape spatially. However, it seems that the mode of complementarity is the most probable and feasible in urban networks, as cities of relatively similar magnitude are well matched and none of them wants to be inferior. The second option of concentration implies the development of a hierarchy in an urban network, which seems contradictory to the historical development pathways such regions have generally followed. Still, however, forces towards the concentration of urban functions in one of the cities of the urban network might be strong, as this might generate more agglomeration economies. Recent indications of such a trend have also become manifest in the classic example of an urban network, the Dutch Randstad. There, it seems that increased specialisation spatially is currently taking place in the Amsterdam region rather than through the mode of complementarity (Van Oort et al., 2006; Lambregts et al., 2006). There is, however, a temporal dimension to this. One may wonder whether the shifting fortunes of the Randstad cities in the past centuries, with one of them growing faster than the others, has finally led to the present overall situation of relatively strong complementarities.

The quest is now for successful regional development strategies that enhance critical mass in practice, whether or not through the mode of complementarity. Stein's Regional City would then perhaps be not such a distant perspective.

References

- ALBRECHTS, L. (2001), 'How to proceed from image and discourse to action: as applied to the Flemish diamond', *Urban Studies*, **38**, 733–45.
- ALONSO, W. (1973), 'Urban zero population growth', *Daedalus*, **109**, 191–206.
- BAILEY, N. and TUROK, I. (2001), 'Central Scotland as a polycentric urban region: useful planning concept or chimera?' *Urban Studies*, **38**, 697–715.
- BATTEN, D. (1995), 'Network cities: creative urban agglomerations for the 21st century', *Urban Studies*, **32**, 313–27.
- BRENNER, N. (2004), *New State Spaces, Urban Governance and the Rescaling of Statehood*, New York and Oxford, Oxford University Press.
- BURTON, I. (1963), 'A restatement of the dispersed city hypothesis', *Annals of the Association of American Geographers*, **63**, 285–89.
- CAMAGNI, R. (1993), 'From city hierarchy to city networks: Reflections about an emerging paradigm', in T. R. Lakshmanan and P. Nijkamp (eds), *Structure and Change in the Space Economy: Festschrift in Honour of Martin Beckmann*, Berlin, Springer Verlag, 66–87.

- CAPELLO, R. (2000), 'The city network paradigm: measuring urban network externalities', *Urban Studies*, **37**, 1925–45.
- CEC (COMMISSION OF THE EUROPEAN COMMUNITIES) (1999), *European Spatial Development Perspective: Towards Balanced and Sustainable Development of the Territory of the EU*, Luxembourg, Office for Official Publications of the European Communities.
- COOMBES, M., CHARLES, D., RAYBOULD, S. and WYMER, C. (2005), 'City regions and polycentricity, the East Midlands urban network' (final report, Newcastle, Centre for Urban and Regional Development Studies).
- DCLG (DEPARTMENT FOR COMMUNITIES AND LOCAL GOVERNMENT) (2006), *Strong and Prosperous communities, The Local Government White Paper* (2 vols), London, DCLG.
- DIELEMAN, F. and FALUDI, A. (1998), 'Polynucleated metropolitan regions in northwest Europe: theme of the special issue', *European Planning Studies*, **6**, 365–77.
- FINCH, D. (2007), 'City-regions: where next?' *Town Planning Review*, **78**, i–vi.
- GOTTMANN, J. (1961), *Megalopolis, The Urbanized Northeastern Seaboard of the United States*, New York, The Twentieth Century Fund.
- GREENACRE, M. J. (1993), *Correspondence Analysis in Practice*, London, Academic Press.
- HAGUE, C. and KIRK, K. (2003), *Polycentricity Scoping Study*, London, Office of the Deputy Prime Minister.
- HARVEY, D. (1989), 'From managerialism to entrepreneurialism: the transformation of urban governance in late capitalism', *Geografiska Annaler*, **71B**, 3–17.
- HM TREASURY, DEPARTMENT FOR BUSINESS ENTERPRISE AND REGULATORY REFORM AND DEPARTMENT FOR COMMUNITIES AND LOCAL GOVERNMENT (2007), *Review of Sub-national Economic Development and Regeneration*, London, HM Treasury.
- HOLCOMB, B. (1994), 'City make-overs: marketing the post-industrial city', in J.R. Gold and S.V. Ward (eds), *Place Promotion: The Use of Publicity and Marketing to Sell Towns and Regions*, Chichester, Wiley.
- KLOOSTERMAN, R. and LAMBREGTS, B. (2001), 'Clustering of economic activities in polycentric urban regions: the case of the Randstad', *Urban Studies*, **38**, 717–32.
- KLOOSTERMAN, R. and MUSTERD, S. (2001), 'The polycentric urban region: towards a research agenda', *Urban Studies*, **38**, 623–33.
- LAMBOOY, J. G. (1969), 'City and city region in the perspective of hierarchy and complementarity', *Tijdschrift voor Economische en Sociale Geografie*, **60**, 141–54.
- LAMBOOY, J. G. (1998), 'Polynucleation and economic development: the Randstad', *European Planning Studies*, **6**, 457–66.
- LAMBREGTS, B. (2000), *Background Study and Theoretical Framework for the EURBANET Project* (EURBANET Report to the European Commission), Delft, OTB Research Institute.
- LAMBREGTS, B., KLOOSTERMAN, R., WERFF, M. VAN DER, RÖLING, R. and KAPOEN, L. (2006), 'Randstad Holland: multiple faces of a polycentric role model', in P. Hall and K. Pain (eds), *The Polycentric Metropolis: Learning from Mega-City Regions in Europe*, London, Earthscan, 137–45.
- LARSEN, K. (2005), 'Cities to come: Clarence Stein's post-war regionalism', *Journal of Planning History*, **4**, 33–51.
- MALECKI, E. J. (2004), 'Jockeying for position: what it means and why it matters to regional development policy when places compete', *Regional Studies*, **38**, 1101–20.

- MARVIN, S., HARDING, A. and ROBSON, B. (2006), *A Framework for City-Regions*, London, Office of the Deputy Prime Minister.
- MEIJERS, E. (2005), 'Polycentric urban regions and the quest for synergy: is a network of cities more than the sum of the parts?' *Urban Studies*, **42**, 765–81.
- MEIJERS, E. (2007a), 'Clones or complements? The division of labour between the main cities of the Randstad, the Flemish Diamond and the RheinRuhr Area', *Regional Studies*, **41**, 889–900.
- MEIJERS, E. (2007b), 'From central place to network model: theory and evidence of a paradigm change', *Tijdschrift voor Economische en Sociale Geografie*, **98**, 245–59.
- MEIJERS, E. (2008), 'Summing small cities does not make a large city: polycentric urban regions and the provision of cultural, leisure and sports amenities', *Urban Studies*, **45**, 2323–42.
- MEIJERS, E. and ROMEIN, A. (2003), 'Realizing potential: building regional organizing capacity in polycentric urban regions', *European Urban and Regional Studies*, **10**, 173–86.
- MINISTERIE VAN VROM (VOLKSHUISVESTING, RUIMTELIJKE ORDENING EN MILIEUBEHEER) (2002), *Vijfde Nota over de Ruimtelijke Ordening 2000/2020, Ruimte maken, ruimte delen, PKB deel 3: kabinetsstandpunt*, The Hague, Ministerie van VROM.
- MINISTERIE VAN VROM (VOLKSHUISVESTING, RUIMTELIJKE ORDENING EN MILIEUBEHEER) (2005), *Nota Ruimte, Ruimte voor Ontwikkeling*, The Hague, Ministerie van VROM.
- NORDREGIO et al. (2004), 'ESPON 1.1.1: Potentials for polycentric development in Europe' (Project report), Stockholm/Luxembourg, Nordregio/ESPON Monitoring Committee.
- NORTH NORTHAMPTONSHIRE JOINT PLANNING UNIT (2008), *North Northamptonshire Core Spatial Strategy*, adopted June 2008, Corby, NNJPU.
- OORT, F. VAN, BRUSSEL, J. VAN, RASPE, O., BURGER, M., DINTEREN, J. VAN, and KNAAP, B. VAN DER (2006), *Economische netwerken in de regio*, Rotterdam/The Hague, NAI Publishers/Ruimtelijk Planbureau.
- PARKINSON, M., et al. (2006), *State of the English Cities* (2 vols), London, ODPM.
- PARR, J. B. (2004), 'The polycentric urban region: a closer inspection', *Regional Studies*, **38**, 231–40.
- PARSONS, K. C. (1998), *The Writings of Clarence S. Stein: Architect of the Planned Community*, Baltimore, The Johns Hopkins University Press.
- PRIEMUS, H. (1994), 'Planning the Randstad: Between economic growth and sustainability', *Urban Studies*, **31**, 509–34.
- STEIN, C. S. (1942), 'City patterns ... past and future', *Pencil Points*, **23**, 52–56.
- STEIN, C. S. (1964), 'A regional pattern for dispersal', *Architectural Record*, 205–06.
- ULLMANN, E. L. (1956), 'The role of transportation and the bases for interaction', in W. L. Thomas (ed), *Man's Role in Changing the Face of Earth*, Chicago, The University of Chicago Press.
- ZONNEVELD, W. (1992), *Naar een beter gebruik van ruimtelijke planconcepten*, Planning Research Report 64, Amsterdam, PDI Universiteit van Amsterdam.
- ZONNEVELD, W. and VERWEST, F. (2005), *Tussen droom en retoriek, de conceptualisering van ruimte in de Nederlandse planning*, Rotterdam/The Hague, NAI Uitgevers/Ruimtelijk Planbureau.

Acknowledgements

The author would like to thank the referees for their useful comments. The author wishes to acknowledge the financial assistance of the Dutch Government through the Habiforum Program Innovative Land Use and Delft University of Technology through the Delft Centre for Sustainable Urban Areas.

Appendix 1. Activities included in the analysis of complementarity

Activity code (NACE Rev. 1.1)	Type of amenity	Code label
5212	Department store	Retail sale in non-specialised stores (with food, beverages or tobacco not predominating)
5510	Hotel	Hotels
5530	Restaurant	Restaurants
5540	Bar	Bars
6330	Travel agency	Activities of travel agencies and tour operators; tourist assistance activities
6512	Bank	Monetary intermediation (not central banking)
6720	Insurance agent, assessor	Activities auxiliary to insurance and pension funding
7411	Legal consulting firm	Legal activities
7450	(Temporary) employment agency	Labour recruitment and provision of personnel
7511	Public services	General (overall) public service activities
8021	Secondary school	General secondary education
8022	Vocational secondary school	Technical and vocational secondary education
8030	Higher education institute	Higher education
8511	Hospital	Hospital activities
8512	Medical practice	Medical practice activities
8515	Hospital	Other hospital activities without overnight stays
9213	Cinema	Motion picture projection
9232	Theatre	Operation of arts facilities
9251	Library	Library and archives activities
9252	Museum	Museums activities and preservation of historical sites and buildings
9261	Sports accommodation	Operation of sports arenas and stadiums
9262	Outdoor sports	Sports outdoor
9263	Indoor sports	Sports indoor
9264	Water sports	Water sports
9265	Other sports	Other sporting activities